

Customer Purchase Prediction Using Logistic Regression

Machine Learning Project Report

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Course : Artificial Intelligence / Machine Learning

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1. AIM

To build a Machine Learning model using Logistic Regression to predict whether a customer will purchase a product based on their Age, Income, and Spending Score.

2. INTRODUCTION

Customer purchase prediction is a Machine Learning problem in which customer information is used to predict whether a customer is likely to purchase a product.

In this project, Logistic Regression is used as the classification algorithm. The model uses Age, Income, and Spending Score as input features and predicts whether the customer will purchase the product or not.

3. DATASET

The dataset contains information about 20 customers. It contains four columns:

- Age – Age of the customer
- Income – Income of the customer
- SpendingScore – Customer's spending score
- Purchased – Indicates whether the customer purchased the product

Purchased is the target variable:

0 = Not Purchased

1 = Purchased

4. DATA EXPLORATION

The dataset was explored using the following Pandas functions:

- `head()` – displays the first five rows of the dataset.
- `shape` – shows the number of rows and columns.
- `isnull()` – checks for missing values.

The dataset contains 20 rows and 4 columns, and there are no missing values.

5. FEATURES AND TARGET

The following features were selected:

- Age
- Income
- SpendingScore

The target variable is:

- Purchased

6. TRAIN-TEST SPLIT

The dataset was divided into training and testing data. 80% of the data was used for training the model and 20% was used for testing.

7. LOGISTIC REGRESSION

Logistic Regression was used to build the classification model.

The model was trained using the training data and then used to predict whether customers in the test dataset would purchase the product.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("\nTraining data size:", X_train.shape)
print("Testing data size:", X_test.shape)

# -----
# 6. Train Logistic Regression model
# -----

model = LogisticRegression()

model.fit(X_train, y_train)
```

8. RESULTS

The trained Logistic Regression model was used to make predictions on the test data.

Accuracy: 100%

```
# -----
# 8. Calculate accuracy
# -----

accuracy = accuracy_score(y_test, y_pred)

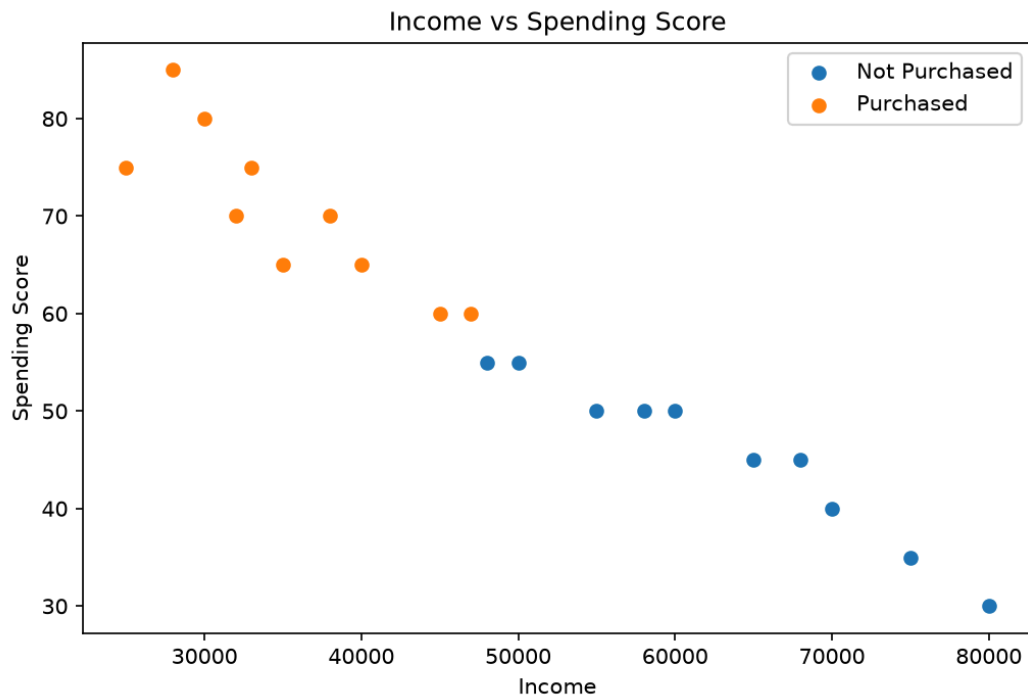
print("\nAccuracy:", accuracy)
print("Accuracy percentage:", accuracy * 100, "%")
```

Accuracy: 1.0

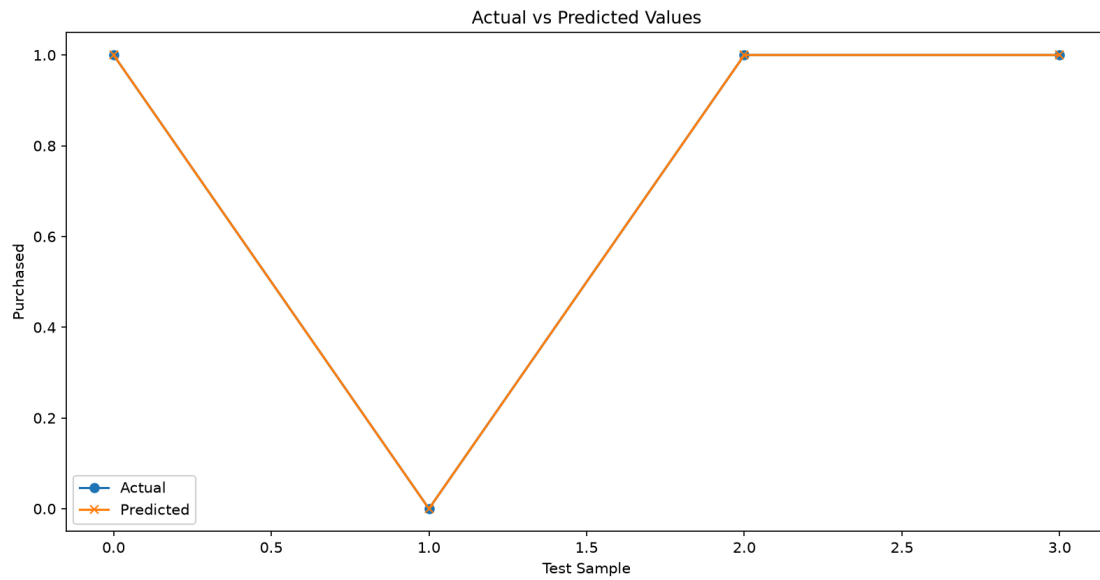
Accuracy percentage: 100.0 %

9. VISUALIZATION

The **Income vs Spending Score** graph shows the distribution of customers who purchased and did not purchase the product.



The Actual vs Predicted graph compares the actual purchase values with the values predicted by the model.



10. NEW CUSTOMER PREDICTION

A new customer's Age, Income, and Spending Score were provided to the trained model to predict whether the customer would purchase the product.

New Customer: Age = 25, Income = 40,000, Spending Score = 70

Actual values:

[1 0 1 1]

Predicted values:

[1 0 1 1]

Accuracy: 1.0

Accuracy percentage: 100.0 %

New Customer Prediction: Customer will purchase. _

11. CONCLUSION

The Logistic Regression model was successfully implemented to predict customer purchase behavior using Age, Income, and Spending Score. The project demonstrates the basic steps of a Machine Learning classification problem, including data exploration, visualization, model training, prediction, and evaluation.